

Year 11
Mathematics Extension 1
Yearly Examination
2011

General Instructions

- Reading time – 5 minutes
- Working time – 1.5 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question

Note: Any time you have remaining should be spent revising your answers.

Total marks – 60

- Attempt Questions 1 – 5
- All questions are of equal value
- Start each question in a new writing booklet
- Write your name on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your name and “N/A” on the front cover

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

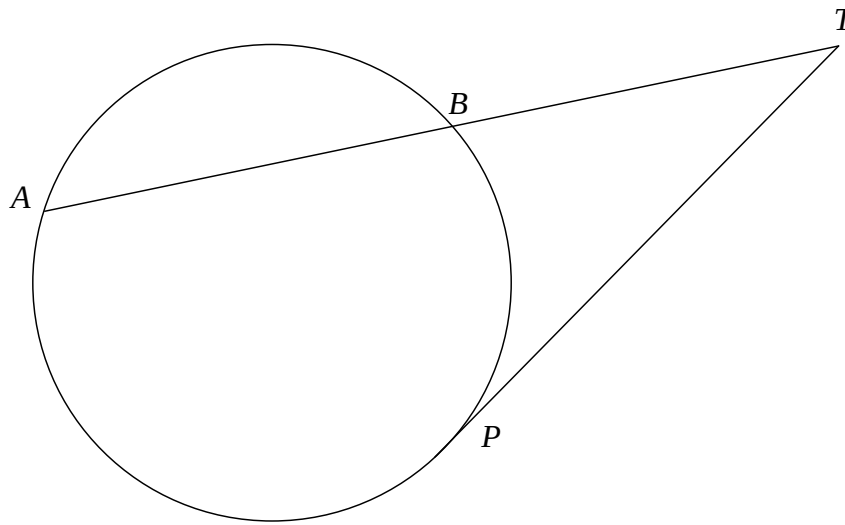
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Total Marks – 60
Attempt Questions 1–5
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks)	Marks
(a) Simplify $\frac{1}{p^2 - pq} - \frac{1}{pq - q^2}$	2
(b) i) Expand $\sin(A - B)$	1
ii) Hence, or otherwise write the exact value of $\sin 15^\circ$	2
(c) Simplify $(n + 1)! - n!$	2
(d) Solve for x where $\frac{x + 2}{x} \geq 0$	3
(e) Write the general solution to the equation $\tan x = 1$	2

(a)



In the diagram PT is a tangent to the circle at P .
 $AB = 5\text{cm}$ and $PT = 8\text{cm}$. Find the length of the interval BT to the nearest tenth of a centimetre.

3

(b) Find the coordinates of the point P which divides the interval AB , where
 $A = (-2, 1)$ and $B = (1, 7)$, internally in the ratio 2:3

2

(c) Find $\lim_{x \rightarrow \infty} \frac{3x + 2x^3}{5x^2 - 1}$

2

(d) Find the numbers of ways in which the letters of the word **FUNCTIONS** can be arranged in a line.

2

(e) i) Sketch $y = |x + 5|$ and $y = 2x$ on the same number plane

1

ii) Find the x coordinate(s) of the point(s) of intersection of the two functions $y = |x + 5|$ and $y = 2x$.

1

ii) Hence, or otherwise solve $2x \leq |x + 5|$

1

Question 3 (12 marks) Use a SEPARATE writing booklet

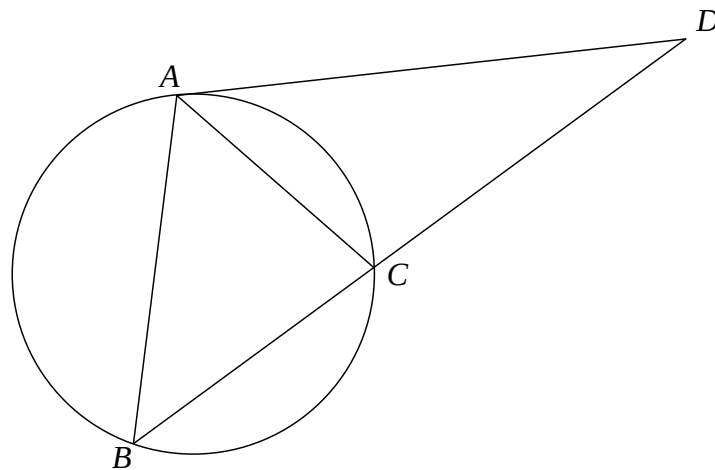
Marks

(a) i) Show that the equation $\cos 2x + 3\cos x + 2 = 0$ is equivalent to the equation $2\cos^2 x + 3\cos x + 1 = 0$ 2

ii) Hence, or otherwise solve the equation $\cos 2x + 3\cos x + 2 = 0$ for $0^\circ \leq x \leq 360^\circ$ 2

(b) Find the domain of the function $f(x) = \sqrt{4 - x^2}$ 2

(c)



ABC is a triangle inscribed in a circle. The tangent to the circle at A meets BC produced at D . $\angle CAB = 60^\circ$, $\angle CDA = 30^\circ$ and $\angle ABC = x^\circ$. Copy the diagram, showing the given information, then find the value of x . 2

(d) In a game of Lotto four numerals are chosen from the numerals **1 2 3 4 5 6 7 8 9** without repetition. Order is not important. Find the number of ways in which this can be done so that

i) **No** odd numerals are chosen 2

ii) **At least** one odd numeral is chosen 2

Question 4 (12 marks) Use a SEPARATE writing booklet

Marks

(a) i) Show that $\cos x - \sqrt{3} \sin x \equiv 2 \cos(x + 60)^\circ$ 2

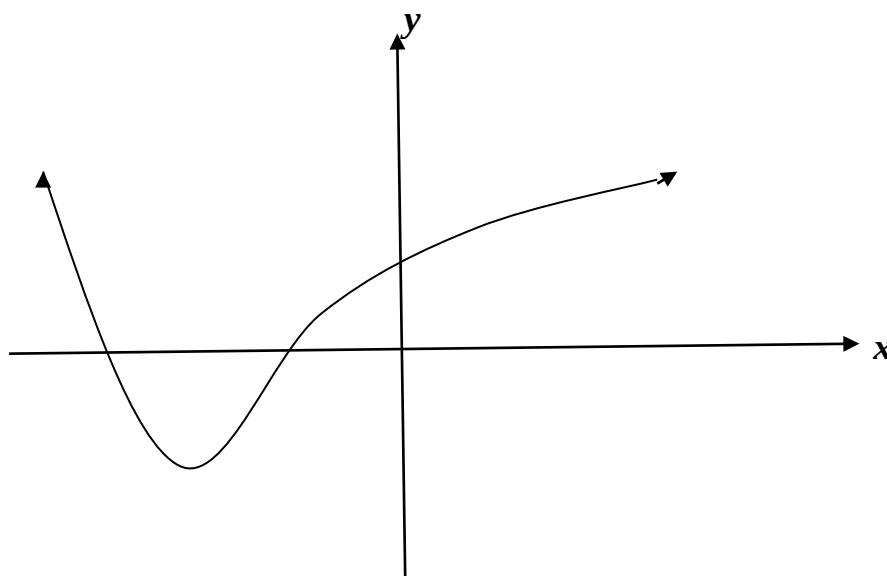
ii) Hence solve $\cos x - \sqrt{3} \sin x = 2$ over $0^\circ \leq x \leq 360^\circ$ 2

(b) i) Show that the acute angle θ between the lines $x - y = 0$ and $x - 2y = 0$ is such that $\tan \theta = \frac{1}{3}$. 2

ii) Find the equation of the other line $y = mx$ which is also inclined at the same acute angle θ to the line $x - 2y = 0$. 2

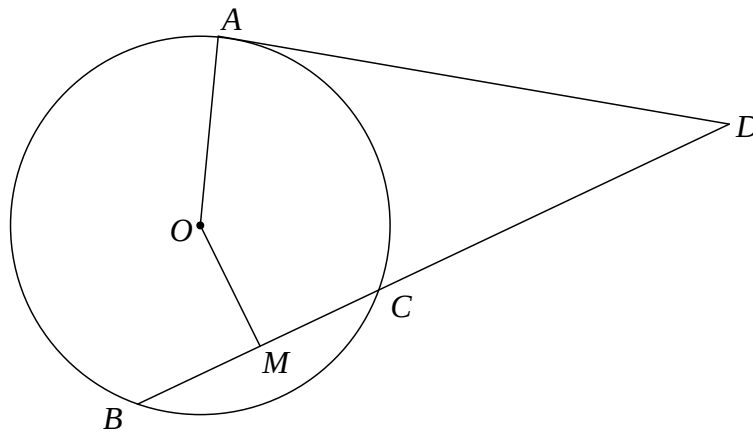
(c) Show that the line $y = x$ is a tangent to the circle $(x - 2)^2 + y^2 = 2$ 2

(d) The graph shown is the function $y = f(x)$. On the answer sheet provided Sketch the gradient function $y = f'(x)$. 2



Marks

- (b)



Copy the diagram into your answer book.

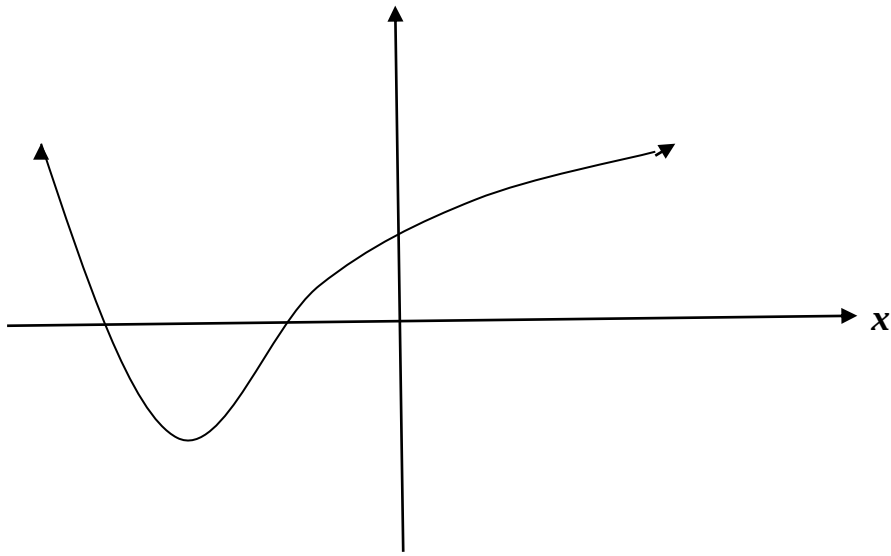
- i) Show that $AOMD$ is a cyclic quadrilateral. 3
- ii) Give a reason why OD is a diameter of the circle through A, O, M and D . 1
- (c) If $t = \tan \frac{x}{2}$ show that $\frac{1 - \cos x}{\sin x} = t$ 2
- (d) 7 boys and 2 girls are to be seated around a circular table. In how many ways can this be done so that the girls are always seated together? 2

End of examination

Answer sheet for Question 4 (d)**Name.....**

The graph shown is the function $y = f(x)$. On the same number plane below sketch the gradient function $y = f'(x)$.

2



Section I:

1. $TA^2 = TC \times TB$

$$12^2 = x \times (x+7)$$

$$144 = x^2 + 7x$$

$$x^2 + 7x - 144 = 0$$

$$(x+16)(x-9) = 0$$

$$x = -16, x = 9$$

$$\therefore x = 9 > 0 \quad (C)$$

2. $1 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40320$ (A)

3. $\sqrt{\frac{4+4(\cos 2x)}{1-\cos 2x}} = \sqrt{\frac{4(1+\cos 2x)}{1-\cos 2x}}$

$$= 2 \sqrt{\frac{1+(2\cos^2 x - 1)}{1-(1+2\sin^2 x)}}$$

$$= 2 \sqrt{\frac{1+2\cos^2 x - 1}{1-1-2\sin^2 x}}$$

$$= 2 \sqrt{\cot^2 x} = 2 \cot x \quad (B)$$

4. $\frac{3 \times (x+4)^2}{x+4} \geq 2 \times (x+4)^2$

$$3(x+4) \geq 2(x+4)^2$$

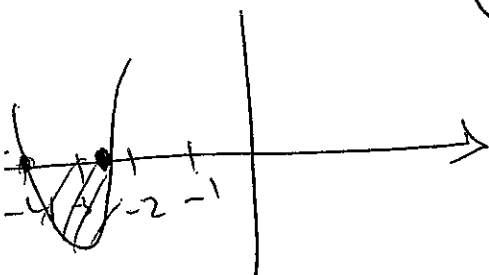
$$2(x+4)^2 - 3(x+4) \leq 0$$

$$(x+4)[2(x+4)-3] \leq 0$$

$$(x+4)(2x+8-3) \leq 0$$

$$(x+4)(2x+5) \leq 0$$

$$-4 \leq x \leq -2\frac{1}{2} \quad (A)$$



5. $\lim_{x \rightarrow \infty} \frac{x+1}{x-1}$

$$= \lim_{x \rightarrow \infty} \frac{\frac{2x^2}{x} + \frac{1}{x}}{\frac{2x}{x} - \frac{1}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{2x}{1} = 2x \quad (C)$$

6. $A(6, -4), B(0, 4), P(-3, 8)$

$$m:n = k:1$$

$$-3 = \frac{k \times 0 + 1 \times 6}{k+1}$$

$$-3k - 3 = 6$$

$$-3k = 9 \rightarrow k = -3 \quad (B)$$

7. $26^2 \times 10^2 \times 26^2 = 45697600$ (D)

8. ${}^9C_5 = 126$ (C)

9. $\lim_{x \rightarrow -1} \frac{x^2 + 2x + 1}{x+1}$
 $= \lim_{x \rightarrow -1} \frac{(x+1)(x+1)}{x+1}$
 $= -1 + 1 = 0 \quad (C)$

10. $\tan 15^\circ = \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ}$

$$= \frac{1 - \frac{1}{\sqrt{3}}}{1 + 1 \times \frac{1}{\sqrt{3}}}$$

$$= \frac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}}$$

$$= \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

$$= \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$$

$$= \frac{3 - 2\sqrt{3} + 1}{3 - 1}$$

$$= \frac{4 - 2\sqrt{3}}{2}$$

$$= 2 - \sqrt{3} \quad (A)$$

Question 11:

a) 2, 3, 5, 6, 8 and 9.

i) $6 \times 5 \times 4 \times 3 \times 2 = 720$ (1)

or

$6P_5 = 720$ 5 digit numbers.

ii. Half the numbers will be even since there's an equal number of odd and even digits.

$720 \div 2 = 360$ (1)

or

$5P_4 \times 3 = 360$

since there are 3 numbers that are even 2, 6 and 8.

b) i. ${}^{16}C_4 = 1820$ (1)

ii. ${}^{10}C_4 = 210$ (1)

iii. At least one man: All combinations - all women

$= {}^{16}C_4 - {}^{10}C_4$
 $= 1610$

* carry thru
mark given
(1) for (i) - (ii)
even if (i) or (ii)
incorrect

or

${}^6C_1 \times {}^{10}C_3 + {}^6C_2 \times {}^{10}C_2 + {}^6C_3 \times {}^{10}C_1 + {}^6C_4 \times {}^{10}C_0 = 1610$

c) i. Four men and 4 women at a circular table:

No restrictions: $7! = 5040$ (1)

ii. 1st position is fixed men: $3! 4! = 144$. (1)

iii. 4 men and one group of women = 5 groups

$\therefore 4!$

4 women = $3!$

2 women can sit together in $2!$ ways

$\therefore$ total number of arrangements = $4! \cdot 3! \cdot 2!$
 $= 288$

-2-

2 Br correct
1 Br dealing
with women
seating.

carry thru
mark given if
(i) answered $4! \times 4!$
and (ii) $4! \times 4! \times 2!$

d) i. $\angle AXC = \angle AYC$

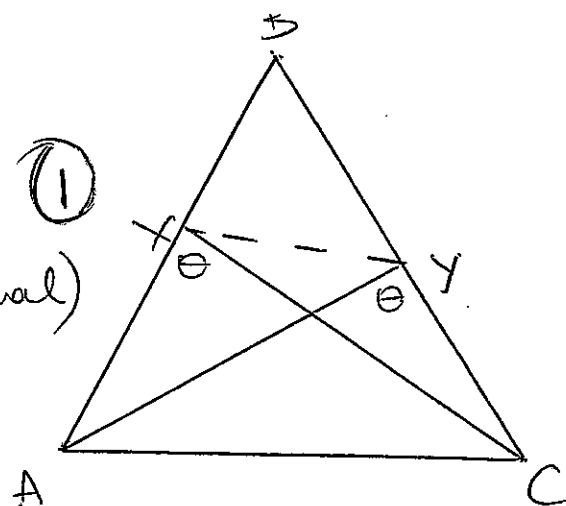
$= \theta$ (given)

$\therefore AXYC$ is cyclic (converse of $\angle$ s in the same segment, AC, are equal) ①

ii. Since $AXYC$ is cyclic,

$\angle BXY = \angle YCA$ (external

$\angle$ in a cyclic quadrilateral = opposite interior $\angle$). ①



e) i. $\cos x - \sin x = A \cos(x + \alpha)$, $0 \leq x \leq 90^\circ$

$A = \sqrt{a^2 + b^2}$

$A = \sqrt{1^2 + 1^2}$

$A = \sqrt{2}$

$\tan \alpha = \frac{b}{a}$

$\tan \alpha = 1$

$\therefore \alpha = 45^\circ$

$\therefore \cos x - \sin x = \sqrt{2} \cos(x + 45^\circ)$

2 mark $\sqrt{2} \cos(x + 45^\circ)$
1 mark either $\sqrt{2}$ or $+45^\circ$.

ii. $\cos x - \sin x = -1$, $0 \leq x \leq 360^\circ$

$\sqrt{2} \cos(x + 45^\circ) = -1$

$\cos(x + 45^\circ) = -\frac{1}{\sqrt{2}}$

2nd and 3rd quadrants.

$\therefore x + 45^\circ = 180^\circ - 45^\circ, 180^\circ + 45^\circ, 45^\circ \leq x + 45^\circ \leq 360^\circ + 45^\circ$

$x + 45^\circ = 135^\circ, 225^\circ$ ①

$x = 90^\circ, 180^\circ$ ① for $0 \leq x \leq 360^\circ$

2 mark for correct answer.

3- 1 mark some correct progress

Question 12:

a) $9^x - 10 \cdot 3^x + 9 = 0$

let $u = 3^x$

then $9^x - 10 \cdot 3^x + 9 = 0$ is equivalent to:

$$(3^x)^2 - 10 \cdot 3^x + 9 = 0$$

$$(3^x)^2 - 10 \cdot 3^x + 9 = 0$$

i.e. $u^2 - 10u + 9 = 0$ - ①

$$(u-9)(u-1) = 0$$

$$u = 9 \text{ or } u = 1$$

$$3^x = 9 \text{ or } 3^x = 1$$

$$x = 2 \text{ or } x = 0$$

①

①

1 for correct u equation
1 for each correct x-value

b) $\therefore \tan 45^\circ = \frac{h}{OC}$ ①

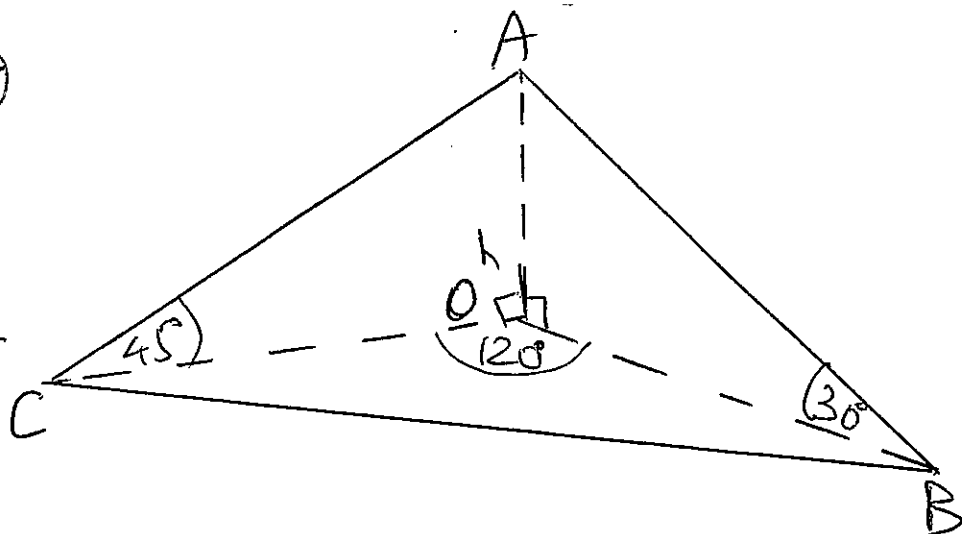
$$1 = \frac{h}{OC}$$

$$\therefore \boxed{OC = h} \quad \text{②}$$

$$\tan 30^\circ = \frac{h}{OB}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{OB}$$

$$\boxed{OB = \sqrt{3}h} \quad \text{③}$$



ii. $BC^2 = OC^2 + OB^2 - 2 \times OC \times OB \times \cos 120^\circ$

$$BC^2 = h^2 + (\sqrt{3}h)^2 - 2 \times h \times \sqrt{3}h \times -\frac{1}{2}$$

1 mark

$$BC^2 = h^2(1 + 3 + \sqrt{3})$$

$$BC^2 = (4 + \sqrt{3})h^2$$

} 1 mark
b.

c) $y = |2x - 3|$

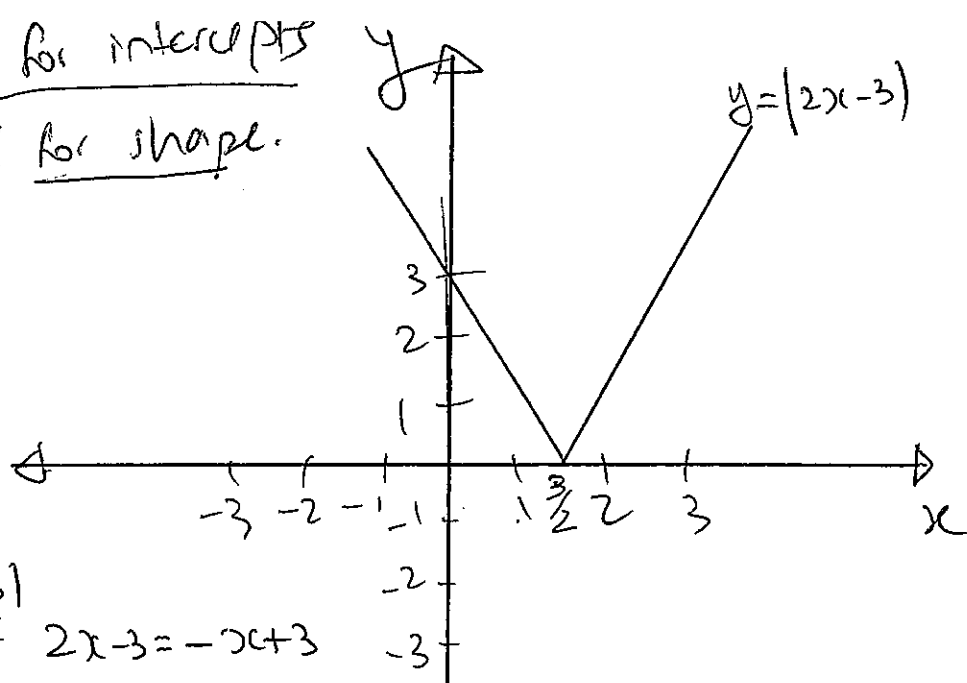
$y = 2x - 3$

or

$y = -2x + 3$

1 for intercepts

1 for shape.



ii. $|2x - 3| \leq |x - 3|$

Solve: $|2x - 3| = |x - 3|$

$2x - 3 = x - 3$ or $2x - 3 = -x + 3$

$x = 0$

or

$3x = 6$

$x = 2$

Test: $x = -1, x = 1, x = 3$

$x = -1$

$|2(-1) - 3| \leq |-1 - 3|$

$|-5| = |-4|$ False.

$x = 1$

$|2(1) - 3| \leq |1 - 3|$

$|-1| \leq |-2|$ True

$x = 3$

$|2(3) - 3| \leq |3 - 3|$

$|3| \leq 0$ False.

$\therefore$ Solution: $0 \leq x \leq 2$

1 for working
1 for result.

d) $A(-4, -3)$, $B(1, 5)$

$m: PA$
 $-3:2$

$$x = \frac{-3 \times 1 + 2 \times -4}{-3 + 2}$$

$$x = \frac{-11}{-1}$$

$$x = 11$$

$$y = \frac{-3 \times 5 + 2 \times -3}{-3 + 2}$$

$$y = \frac{-15 - 6}{-1}$$

$$y = 21 \quad 1 \text{ for each coordinate}$$

$\therefore P(11, 21)$

1 mark for significant progress towards answer

e) $\frac{2x+5}{x+1} < 1 \quad x(x+1)^2, x \neq -1$

$$(2x+5)(x+1) < (x+1)^2$$

$$(x+1)^2 - (2x+5)(x+1) > 0$$

$$(x+1)[(x+1) - (2x+5)] > 0$$

$$(x+1)(x+1-2x-5) > 0$$

$$(x+1)(-x-4) > 0$$

$$\boxed{-4 < x < -1}$$

or

Solve $\frac{2x+5}{x+1} = 1$ then test x -values:

$$2x+5 = x+1$$

$$x = -4$$

Test: $x = -5, x = -2, x = 0$

$x = -5$:

$$\frac{2x-5+5}{-5+1} < 1$$

$$\frac{-5}{-4}$$

$$\frac{-5}{-4} < 1 \text{ False.}$$

$$\frac{-5}{-4}$$

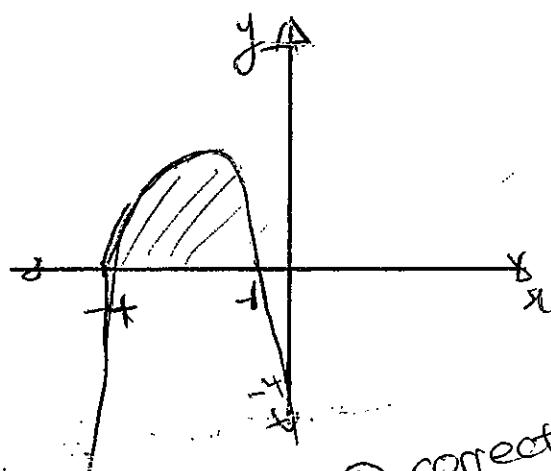
$x = -2$

$$\frac{2x-2+5}{-2+1} < 1$$

$$\frac{1}{-1}$$

$$\frac{1}{-1} < 1 \text{ True.}$$

$$\frac{1}{-2}$$



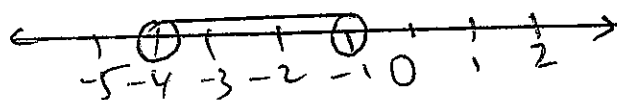
① correct progress having noted $x \neq -1$

$x = 0$

$$\frac{2x+5}{1} < 1$$

$$5$$

$$5 < 1 \text{ False.}$$



$$\therefore -4 < x < -1$$

Question 13:

a) i. $y = x^2 + 2x - 3$ and $y = x^2 + 1$

$$x^2 + 2x - 3 = x^2 + 1$$

$$2x = 4$$

$$x = 2$$

solving simultaneously.

$$y = x^2 + 1 \rightarrow y = 2^2 + 1 \rightarrow y = 5$$

any correct working.

$\therefore (2, 5)$

ii. $y = x^2 + 2x - 3$

$$y' = 2x + 2$$

$$\therefore m_1 = 2 \times 2 + 2$$

$$\boxed{m_1 = 6}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = \left| \frac{6 - 4}{1 + 6 \times 4} \right|$$

$$\tan \theta = \left| \frac{2}{25} \right|$$

$$\therefore \theta = 4.5739 \dots$$

$$\therefore \theta \doteq 5^\circ$$

$y = x^2 + 1$
 $y' = 2x$

$$\therefore m_2 = 2 \times 2$$

$$\boxed{m_2 = 4}$$

1 for gradients

1 for result.

b) $4x^2 - (6+k)x + 1 = 0$

No real zeros $\rightarrow \Delta < 0$

$$b^2 - 4ac < 0$$

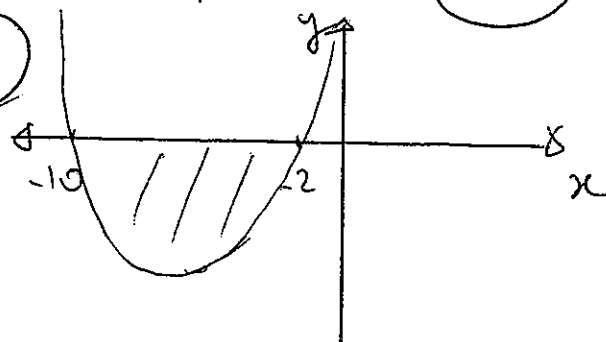
$$[-(6+k)]^2 - 4 \times 4 \times 1 < 0 \quad \textcircled{1}$$

$$36 + 12k + k^2 - 16 < 0$$

$$k^2 + 12k + 20 < 0$$

$$(k+2)(k+10) < 0$$

$$\therefore -10 < k < -2 \quad \textcircled{1}$$



c) i. $x \neq \pm 3$ ① $f(x) = \frac{2x^2}{x^2-9}$
 $x=3$ or $x=-3$

ii. $\lim_{x \rightarrow \pm\infty} \frac{2x^2}{x^2 - \frac{9}{x^2}}$

$\lim_{x \rightarrow \pm\infty} \frac{2}{1-0}$

$=2$, y approaches 2 as $x \rightarrow \pm\infty$ ①

iv. Range: $y \leq 0, y > 2$

d) $y = mx$
 $2x - y - 1 = 0$
 $y = 2x - 1$
 $\therefore m_1 = 2$

$\tan 45^\circ = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$1 = \left| \frac{2 - m}{1 + 2m} \right|$

$\therefore 1 = \frac{2 - m}{1 + 2m}$ or $-1 = \frac{2 - m}{1 + 2m}$

$1 + 2m = 2 - m$

$-1 - 2m = 2 - m$

$3m = 1$

$-m = 3$

$\therefore m = \frac{1}{3}$ ①

$\therefore m = -3$ ①

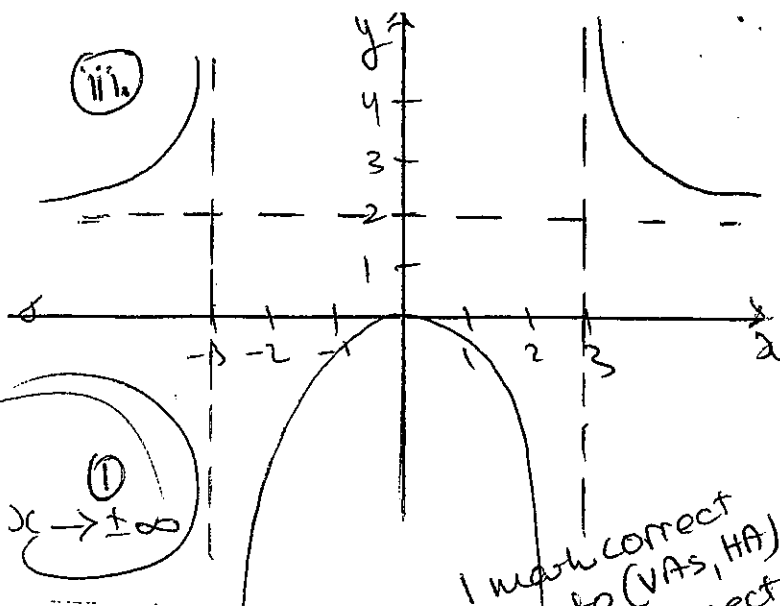
e) $\cos 2x - 3\cos x + 2 = 0$
 $2\cos^2 x - 1 - 3\cos x + 2 = 0$ ←
 $2\cos^2 x - 3\cos x + 1 = 0$
 $(2\cos x - 1)(\cos x - 1) = 0$
 $2\cos x - 1 = 0, \cos x - 1 = 0$

$2\cos x = 1$ or $\cos x = 1$

$\cos x = \frac{1}{2}$

$x = 60^\circ, 360^\circ - 60^\circ$ or $x = 0^\circ, 360^\circ$

$\therefore x = 0^\circ, 60^\circ, 300^\circ, 360^\circ$



1 mark correct into (VAs, HA)
 1 mark correct graph.

1 mark may be awarded for correct progress, eg: correct substitution or one correct solution.

1 for correct substitution for $\cos 2x$

1 for correct $\cos x$ values

1 for solution.